

PROFILE

Computational cognitive scientist and machine learning researcher with 5+ years of experience designing rigorous evaluations, behavioral studies, and reproducible analysis pipelines across child development, adolescent decision-making, and human-centered AI systems. Expertise in combining quantitative methods in developmental psychology, cognitive neuroscience, and software engineering to study complex behaviors. Recent work spans frontier language model evaluation, developmental research using multimodal human-subjects data, and open-science programs for improving research quality.

EDUCATION

University of Edinburgh , Edinburgh, UK	09/2022 – 10/2025
Doctoral-track Candidate in Psychology	09/2023 – 10/2025
Project: <i>Structural and Functional Neural Mechanisms of Theory of Mind Development</i>	
M.Sc. in Psychological Research	Awarded With Merit, 11/2025
Dissertation: <i>Studying the distinct functions of theory of mind brain regions during movie-viewing</i>	
UKRI ESRC DTP Studentship	09/2022 – 10/2025
Scripps College , Claremont, USA	08/2017 – 05/2021
B.A. in Computer Science (Harvey Mudd College) and Neuroscience (Keck Science)	Awarded Cum Laude, 05/2021
Thesis: <i>Uncovering Object Categories in Infant Views</i>	
Scripps College Humanities Institute Fellowship	01/2021
Scripps College Success Grant Scholarship	08/2020, 08/2021
Stanford University NSF REU Scholarship at Center for the Study of Language and Information	02/2020
University of Minnesota NSF REU Scholarship in Neuroimaging for Cognitive Neuroscience	05/2019

PROFESSIONAL & RESEARCH EXPERIENCE

AI Expert Evaluator, Outlier & Alignerr	10/2025 – Present
- Evaluating frontier language models on reasoning and coding tasks, designing and using rubrics to identify failure modes. - Comparing model judgments across ambiguous prompts to surface alignment gaps, especially regarding reproducibility.	
Guest Lecturer, Bluebridge Education	03/2024
- Designed and delivered two interdisciplinary lectures bridging expertise in cognitive neuroscience and AI for advanced secondary-school students. <i>Brain-Inspired Artificial Intelligence</i>: led discussion-based lesson on the neurobiological visual system (covering receptive fields, Hubel & Wiesel, ventral pathway), and mapped those mechanisms to the history of neural networks (i.e., McCulloch-Pitt threshold units, Hebbian learning, Rosenblatt/multi-layer perceptrons, and convolution). <i>Applications of AI: Cognitive Science</i>: taught interdisciplinary foundations of cognitive science (e.g., brain versus mind), introduced functional MRI methodologies, and connected neuroimaging data to machine learning tools, such as multi-voxel pattern analysis and support vector machines.	
Postgraduate Researcher, University of Edinburgh	09/2023 – 10/2025
- Designed analyses for studying theory of mind development in children aged 5 to 12, integrating structural and functional MRI with behavioral outcome measurements. - Built reproducible Python-based analysis pipelines for processing and analyzing functional and diffusion MRI data, paying special attention to anonymization to protect data from child participants. - Led workshops and hackathons to align researchers, clinicians, and technical contributors across concurrent projects, teaching data processing, organization, and analysis for studying cognitive development.	
Open Research Program Lead, PPLS Open Science, University of Edinburgh	08/2023 – 10/2025
- Designed and scaled a program for researchers across 3 departments to review code for publication-ready projects. - Advised researchers on open science practices at all stages of the research lifecycle, hosting workshops and 1:1 consultations on data governance, preregistration, and reproducible research design.	
Junior Research Associate, New York University	05/2021 – 08/2022
- Managed IRB/ethics compliance, budgeting, and codebase for high-impact research on adolescent decision-making, reinforcement learning, and mental health risk factors. - Assured high quality collection, processing, and analysis workflows for multimodal data, including online experiments, eye-tracking, and functional MRI.	

- Led school outreach and participant recruitment with NYC public and private schools, coordinating with educators and caregivers to support safe, informed participation of minors.

Software Engineer, **Stanford University**

05/2021 – 11/2021

- Engineered a participant database system with custom Python tooling for all lab participants, improving scalability and coordinating overhead in human-subjects research operations.

Undergraduate Thesis Researcher, **Scripps College & Stanford University**

08/2020 – 05/2021

- Built computer vision pipelines for infant egocentric video using Detectron2 models to characterize early visual experience and relate naturalistic input to object and word learning.

Data Curator, **Peekbank Project**

05/2020 – 08/2021

- Streamlined integration of child development datasets from 12+ research labs into a shared open-access repository, enabling large-scale cross-study analyses of early language and attention.

Summer Research Intern, **Stanford University**

05/2020 – 08/2020

- Built human-in-the-loop scalable annotation workflows with AWS tooling to label egocentric videos collected in infants' home environments.

Summer Research Intern, **University of Minnesota**

05/2019 – 08/2019

- Implemented supervised machine learning models to successfully predict perceived color using functional MRI data.

SELECTED PUBLICATIONS

- [1] Novelty and uncertainty differentially drive exploration across development
eLife, 2023
Kate Nussenbaum, Rebecca E. Martin, Sean Maulhardt, Yi Yang, Greer Bizzell-Hatcher, **Naiti S. Bhatt**, Maximilian Scheuplein, Gail M. Rosenbaum, John P. O'Doherty, Jeffrey Cockburn, & Catherine A. Hartley
- [2] Peekbank: Exploring children's word recognition through an open, large-scale repository for developmental eye-tracking data
Behavior Research Methods, 2022
Martin Zettersten, ..., **Naiti S. Bhatt**, Claire A. Bergey, & Michael C. Frank
- [3] Characterizing the object categories two children see and interact with in a dense dataset of naturalistic visual experience
Proceedings of the 43rd Annual Conference of the Cognitive Science Society, 2021
Bria Long, George Kachergis, **Naiti S. Bhatt**, & Michael C. Frank
- [4] Classification Analyses of fMRI Data Predict Perceived Color
Poster presented at Southern California Conference for Undergraduate Research (SCCUR), 2019
Poster presented at Bay Area Vision Research Day (BAVRD), 2019
Naiti S. Bhatt, Katherine E.M. Tregillus, & Stephen A. Engel

LEADERSHIP & COMMUNITY

Postgraduate Student Representative in Psychology, **University of Edinburgh**

09/2023 – 10/2025

- Elected by 30+ postgraduate researchers to represent student interests in faculty board meetings and university-level committees; successfully advocated to preserve community funding amidst university-wide budget cuts.

Pre-Conference Workshop Facilitator, **Flux Society (UCL & NYU)**

09/2021

- Designed and presented an invited workshop on computational modeling for cognitive neuroscience for attendees at an international conference.

STEAM Coordinator, **Scripps College Academy**

01/2021 – 05/2021

- Founded and led a program serving underrepresented high school students across Los Angeles and the Inland Empire.

Conference Volunteer, **Women in Machine Learning (NeurIPS)**

12/2019, 12/2021

- Supported community-building and logistics for workshop attendees within the broader NeurIPS research community.

ADDITIONAL SKILLS

Methods & Evaluation: Experimental design, AI evaluation and benchmarking, human-subjects research, preregistration, ethics protocols, open science, data governance.

Programming: Python, R, Linux, Git, TensorFlow, JavaScript (jsPsych), Detectron2, Matlab, React, Docker, HPC/SLURM.

Domains: Child development, reinforcement learning, neuroimaging, naturalistic behavioral data, reproducible research.